

EV Connect AI

Pan-India Intelligent EV Charging Marketplace, Real-Time OCPP Telemetry, Atomic Slot Engine & AI Route Optimization Architecture

PLATFORM METADATA & REPOSITORY SPECIFICATION

Author & Engineering Lead:	EV Connect Core Team
Software Release:	v2.0.0 (Fullstack Production)
Target Network:	14 Indian Metropolitan Hubs & Expressways
GitHub Repository:	DaddyYoda7/Ev-Charging-Marketplace-development-2.0
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1. Executive Summary & Core Mission

EV Connect AI is an enterprise-grade, two-sided electric mobility platform engineered specifically for the Indian charging grid. It bridges the infrastructure gap between EV Drivers, Station Hosts / Charge Point Operators (CPOs), and Grid Administrators by delivering unified search across 14 cities, atomic double-booking prevention, dynamic time-of-day tariffs, simulated UPI/Card payments with GST tax invoices, and real-time OCPP 1.6-J / 2.0.1 telemetry.

Stakeholder Persona	Core Platform Capabilities	Value Proposition & Impact
EV Drivers (2W & 4W)	Pan-India discovery, 45-min slot reservation, garage manager with SoC sliders, simulated payments, AI highway trip planner.	Eliminates range anxiety, guarantees slot availability, and supports both 2W battery swaps & 4W Ultra-Fast DC charging.
Station Hosts (CPOs & Owners)	Station builder, bay hardware management (3.3kW to 350kW), dynamic kWh tariff editor, and 90% payout settlement.	Maximizes bay utilization, monetizes peak traffic hours, and automates financial reconciliations.
Platform Admins (Treasury & Grid)	National grid telemetry hub, 10% platform commission ledger, station verification audit, and predictive failure diagnostics.	Full visibility over national EV infrastructure health, Gross Merchandise Value (GMV), and dispute management.

2. Technology Stack & Architectural Tooling

Layer	Technology / Tool	Version	Architectural Role & Functionality
Frontend	React	^19.2.8	Component-based UI framework managing modals, state, and telemetry animations.
Frontend	Vite	^8.2.0	Next-gen build tool & dev server with instant Hot Module Replacement (HMR).
Frontend	Tailwind CSS	^4.0	Utility CSS engine customized with glassmorphism and backdrop blurs.
Frontend	Leaflet Maps	^1.9.4	Interactive map engine with CartoDB dark tiles, 2W/4W pins, and polyline routes.
Frontend	Lucide React	^1.33.0	Modern SVG icon system for UI controls, navigation, and badges.
Backend	Node.js	>=18.0.0	Asynchronous non-blocking runtime powering server APIs and event streams.
Backend	Express.js	^4.21.2	RESTful routing framework, error middleware, and production static server.
Database	SQLite3	^5.1.7	ACID-compliant transactional relational database engine.
Real-Time	Server-Sent Events	Native HTTP	Unidirectional real-time stream pushing live power, voltage, current, and SoC %.

3. Detailed Phased Roadmap Implementation (Phases 1 – 6)

Phase 1: User & EV Driver Ecosystem

- Multi-Role Switcher: Instant switching between Driver (Cyan), Host (Emerald), and Admin (Purple).
- EV Garage Manager: Vehicle selector with real-time battery SoC sliders supporting Ather, Ola, Tata, MG, and Hyundai.
- Pan-India Interactive Map: Leaflet engine plotting 15 charging hubs across 14 Tier-1/2 Indian cities with city quick-jumps.
- Dynamic Filter Grid: Search bar, category filters (All, 2W Scooty, 4W Fast DC), transparent frosted connector dropdown with full background blur, max tariff slider, and Open Bays toggle.

Phase 2: Station Owner / Host Marketplace

- Station Builder: Registration form wizard for physical hubs with GPS coordinates, address, and amenities (WiFi, Café, CCTV).
- Bay Hardware Configurator: Manage individual chargers from 3.3kW AC (15A socket) up to 350kW DC Ultra-Fast.
- Dynamic Tariff Engine: Configure base rates (₹/kWh), peak-hour surcharges (1.25x), and idle parking penalties (₹/min).

Phase 3: Slot Reservation & Conflict Prevention

- 45-Minute Slot Matrix: Dynamic time slot generation for today and tomorrow based on station operational hours.
- Atomic Concurrency Protection: Database-level overlap validation ensuring no bay can be double-booked simultaneously.
- Lifecycle State Machine: CONFIRMED -> ACTIVE (charging in progress) -> COMPLETED or CANCELLED.

Phase 4: Financial Transactions, Revenue Splits & Invoicing

- Multi-Method Checkout: Simulated payment gateway supporting UPI QR (GPay, PhonePe), Cards, and EV Connect Digital Wallet.
- Automated 10/90 Revenue Split: 10% platform commission retained in treasury, 90% payout settled to station host.
- Cryptographic GST Tax Invoices: Downloadable/printable tax invoices signed with unique HMAC-SHA256 verification hashes.

Phase 5: Multi-Factor AI Recommendation Engine

- Mathematical Scoring Model: $\text{Score} = (0.30 * \text{Availability}) + (0.25 * \text{Proximity}) + (0.20 * \text{Price}) + (0.15 * \text{Speed}) + (0.10 * \text{Rating})$.
- Explainable AI (XAI) Badges: Natural language decision reasoning tags (e.g. 'Top Pick: 60kW Fast DC + 100% Green Solar').
- AI Highway Route & Battery Planner: Computes trip distance, estimated kWh consumption, remaining SoC %, and recommends optimal charging stops.

Phase 6: Real-Time OCPP 2.0.1 & Predictive Maintenance

- OCPP Protocol Simulation: Implements BootNotification, StatusNotification, Heartbeat, MeterValues, StartTransaction, and StopTransaction.
- Fault Injection Sandbox: Live simulation of Overheating, Ground Fault, and Connector Lock Failure conditions.
- Predictive Maintenance (MTBF): Evaluates voltage variance, connector cycles, and temperature to calculate Mean Time Between Failures.

4. Backend Database Schema & REST API Specifications

Primary Database Tables (SQLite3):



Key REST API Endpoints:

Endpoint	Method	Description & Payload Details
/api/auth/users	GET	Fetch registered stakeholder profiles and wallet balance.
/api/auth/vehicles/:id/soc	PUT	Update live EV battery state of charge (SoC %).
/api/stations	GET	Filter stations by keyword, connector, power, price, and geo-radius.
/api/stations/:id/tariff	PUT	Update base tariffs, peak multipliers, and idle parking fees.
/api/bookings/slots	GET	Generate 45-minute available slot matrix for today and tomorrow.
/api/bookings	POST	Reserve a charging bay with atomic concurrency conflict check.
/api/payments/process	POST	Execute payment with 10% platform / 90% host revenue split.
/api/payments/invoice/:id	GET	Generate cryptographic GST tax invoice with HMAC signature.
/api/ai/recommend	POST	Compute multi-factor recommendation score for vehicle & location.
/api/ai/trip-planner	POST	Plan long-distance EV route with battery discharge curves.
/api/telemetry/stream	GET	Persistent Server-Sent Events (SSE) live telemetry stream.
/api/telemetry/ocpp/event	POST	Dispatch simulated OCPP hardware events (Start/Stop/Fault).

5. Local Execution & Cloud Deployment Guide

Local Development (1-Click Launch):

Double-click start.bat in the project root folder. It starts both the Express server on port 5000 and the Vite frontend dev server on port 5173.



Production Cloud Deployment Options:

Provider	Type	Deployment Steps
Render.com	Fullstack (Node + SQLite)	Connect repository DaddyYoda7/Ev-Charging-Marketplace-development-2.0. Build Command: cd client && npm install && npm run build Start Command: cd server && npm install && node index.js
Railway.app	Fullstack Container	Deploy directly from GitHub repository and generate a public domain under service settings.
Vercel	Frontend Client (SPA)	Import repository → Framework: Vite → Root Directory: client.